

Artificial Intelligence in Ethiopian Higher Education: A Bibliometric Analysis of Research Trends, Systemic Adoption Barriers, and Policy-Infrastructure Disconnects

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Abstract

The rapid ascent of artificial intelligence is transforming higher education globally, yet adoption in Sub-Saharan Africa remains critically understudied. Ethiopia presents a compelling paradox: its ambitious National Artificial Intelligence Policy (2024) positions AI as central to national development, yet the country ranks among the world's lowest in AI readiness, creating a profound mismatch between policy vision and infrastructural reality. This study systematically investigates AI research trends, systemic adoption barriers, and policy-implementation disconnects in Ethiopian higher education using a mixed-methods bibliometric-synthesis design. We analyzed academic publications indexed in Scopus and Web of Science (2000–2026) combined with qualitative thematic analysis of full-text articles and policy documents, employing VOSviewer and Biblioshiny for bibliometric mapping and inductive thematic coding across infrastructure, pedagogy, language, and human-capacity dimensions. Results reveal that Ethiopian AI scholarly output has sharply increased after 2023, indicating nascent scholarly engagement rather than systemic adoption. Applying the UTAUT framework, we identify an "adoption paradox": high performance expectancy coexists with critically weak facilitating conditions; unreliable electricity, limited connectivity, prohibitive hardware costs (2–3× global prices), and inadequate teacher training. Linguistic challenges are prominent, with 55.6% of students demanding local language support. We demonstrate significant disconnects between the National AI Policy and institutional realities across infrastructure, data management, and ethics governance. We conclude that adoption barriers are rooted in systemic socio-economic factors and implementation failures, risk exacerbating the "Matthew Effect" by advantaging well-resourced urban universities. We provide policy recommendations and propose an extended UTAUT framework incorporating infrastructure and linguistic capacity for Global South higher education contexts.

Keywords

artificial intelligence in education; bibliometric analysis; Ethiopian higher education; policy-implementation gaps; Global South



I. Introduction

1.1 The Global Imperative of AI in Higher Education

Artificial intelligence has emerged as a transformative force in higher education globally, fundamentally reshaping pedagogical paradigms, assessment methodologies, and student support systems. Since McCarthy's conceptualization of AI in 1956, the field has undergone remarkable evolution, now offering diverse applications including personalized learning pathways, automated assessment systems, and predictive analytics for student success (Ahsina et al., 2025). A comprehensive bibliometric analysis of 1,780 articles published between 1986 and 2025 reveals three distinct phases in AI's integration into higher education: a latency phase (1986–2013) characterized by expert systems and rule-

based approaches; a moderate growth phase (2014–2019) marked by the emergence of chatbots and adaptive learning technologies; and a post-2020 explosion driven by generative models such as ChatGPT (Ahsina et al., 2025).

This exponential growth signifies a fundamental shift in scholarly discourse from debating whether AI should be integrated into pedagogy to examining how it should be integrated effectively, ethically, and equitably. Recent bibliometric studies confirm that AI is rapidly transforming teaching and learning processes across higher education institutions worldwide (Oguntona & Emere, 2025; Goshu, 2026). However, the geographical concentration of AI research remains pronounced, with most publications originating from the United States and China, while critical themes related to accessibility, ethics, and frugal AI remain scarcely addressed (Ahsina et al., 2025). This geographical and thematic concentration underscores a critical gap: the Global South, and Sub-Saharan Africa in particular, remains severely underrepresented in the global AI-in-education research landscape (Oguntona & Emere, 2025; Goshu, 2026).

1.2 The Ethiopian Context: Ambition vs. Reality

Ethiopia presents a compelling paradox in the global AI discourse, a nation demonstrating ambitious policy commitment to digital transformation while confronting severe infrastructural and capacity constraints. The "Digital Ethiopia 2025" strategy established a foundational framework for transforming technical and professional sectors, successfully digitalizing over 900 government services in alignment with the country's Homegrown Economic Reform Agenda and the African Union's Digital Transformation Strategy (Ministry of Innovation and Technology, n.d.).

The most significant policy milestone occurred in June 2024 when Ethiopia's Council of Ministers approved the country's first National Artificial Intelligence Policy (Federal Democratic Republic of Ethiopia, 2024). This strategic document provides a national framework to guide AI development, adoption, and governance, with priority actions focusing on expanding AI education and training, strengthening lifelong learning, and developing leadership and professional skills (OECD.AI Policy Observatory, 2026).

Yet this policy ambition collides with stark reality. The World Bank's 2025 Digital Progress and Trends Report places Ethiopia among the world's lowest-performing countries in AI readiness, classifying it as a low-income nation lacking both the infrastructure to develop AI technologies and the per-capita capacity to adopt them across public services and daily life (World Bank, 2025; The Reporter Ethiopia, 2025). Ethiopia's small, fragmented digital market, combined with limited computing infrastructure and acute skills shortages, places it at risk of becoming a permanent consumer of foreign AI platforms rather than a creator of locally suited technologies (World Bank, 2025).

The infrastructure deficit is multidimensional. More than 60% of developing countries face serious energy security challenges, while Ethiopia confronts a scarcity of advanced digital and AI skills compounded by under-resourced educational institutions and ongoing brain drain (The Reporter Ethiopia, 2025). Language and data gaps present another critical barrier, as many local languages remain underrepresented in global AI training data, risking the exclusion of large population segments from AI's benefits (World Bank, 2025). The World Bank warns that without dedicated investments in local language models and culturally relevant interfaces, AI risks replicating and reinforcing existing inequities (The Reporter Ethiopia, 2025).

1.3 Problem Statement and Research Gap

While global bibliometric studies on AI in education have proliferated, specific, synthesized evidence for Sub-Saharan Africa, and Ethiopia specifically, remains critically lacking. A bibliometric analysis of AI applications for teaching and learning within African higher education institutions reveals a sharp increase in AI-related publications since 2023, with South Africa, Nigeria, and Morocco emerging as key contributors (Oguntona & Emere, 2025). Ethiopia's presence in this literature, however, remains fragmented and peripheral.

Existing Ethiopian studies are characterized by methodological and thematic fragmentation. Research has focused predominantly on technology adoption models using the Unified Theory of Acceptance and Use of Technology (UTAUT) framework, teacher perceptions, and single-institution case studies. A study investigating Ethiopian university students' behavioral intentions and AI tool usage through an extended UTAUT model confirmed that AI application in resource-limited countries like Ethiopia remains very low (Krishnan et al., 2025; Goshu and Woldeamanual, 2026). However, no comprehensive study has systematically mapped the bibliometric landscape of AI research in Ethiopian higher education, identified systemic adoption barriers across infrastructural, pedagogical, linguistic, and human-capital dimensions, or critically examined alignment between national policy frameworks and ground-level institutional realities. This fragmented evidence base impedes evidence-informed policymaking and strategic investment in AI for Ethiopian higher education.

1.4 Research Questions

This study addresses the identified gaps through the following research questions:

RQ1: What are the bibliometric characteristics, including publication growth trajectories, key authors, institutional contributions, and international collaboration patterns of AI research in Ethiopian higher education?

RQ2: What are the primary systemic barriers (infrastructural, pedagogical, linguistic, and human-capital) hindering AI adoptions in Ethiopian pedagogy?

RQ3: To what extent does the current national policy framework align with or disconnect from ground-level infrastructural and institutional realities?

This study makes several contributions to theory, policy, and practice. Theoretically, it provides the first comprehensive bibliometric mapping of AI research in Ethiopian higher education, establishing a baseline for future longitudinal analysis and offering a contextualized extension of technology acceptance frameworks for resource-constrained settings. For policymakers—including Ethiopia's Ministry of Education and the Ethiopian Artificial Intelligence Institute, the study provides evidence-based insights to inform implementation of the National AI Policy's education priorities. For university administrators, the findings offer a diagnostic framework for assessing institutional AI readiness..

II. Review of Literature

2.1 Theoretical Underpinning: The Unified Theory of Acceptance and Use of Technology (UTAUT)

The Unified Theory of Acceptance and Use of Technology (UTAUT), developed by Venkatesh and colleagues (2003), provide a robust theoretical lens for examining technology adoption in educational contexts. The model synthesizes eight prominent technology acceptance theories and identifies four core constructs that determine behavioral intention and actual usage behavior. Performance Expectancy refers to the degree to which an individual believes that using the technology will enhance job performance. Effort Expectancy captures the perceived ease of use associated with the technology. Social Influence reflects the extent to which individuals perceive that important others believe they should use the technology. Facilitating Conditions encompass the objective environmental factors that enable technology use, such as infrastructure, technical support, and training resources (Venkatesh et al., 2003). These constructs are moderated by gender, age, experience, and voluntariness of use.

The UTAUT model has been extensively applied in educational technology research across diverse contexts. Bayaga and Madimabe (2024) employed the UTAUT model to examine teacher collaboration in the indigenization of mathematics pedagogy in South African higher education, demonstrating the model's utility in understanding technology-mediated collaborative practices in resource-constrained settings. Similarly, studies investigating AI-driven virtual learning environments in African higher education institutions have utilized the UTAUT model alongside Diffusion of Innovations theory and the Theory of Planned Behavior to provide valuable frameworks for understanding AI adoption patterns (Potential of AI-Driven Virtual Learning Environments in African Higher Education Institutions, 2025).

The choice of UTAUT as the analytical lens for this study is particularly justified by its emphasis on Facilitating Conditions, which encompass the infrastructural and technical support dimensions that are critically precarious in resource-constrained settings such as Ethiopia. In contexts where reliable electricity, internet connectivity, and technical support are not guaranteed, the facilitating conditions construct becomes a decisive factor in technology adoption outcomes.

2.2 AI in Higher Education: A Global-South Perspective

The integration of artificial intelligence in higher education across Sub-Saharan Africa presents a complex landscape characterized by both transformative potential and formidable obstacles. A scoping review investigating AI's role within higher education institutions in Sub-Saharan Africa underscores the potential of AI to substantially improve educational quality through personalized learning experiences and enhanced operational efficiencies, while simultaneously identifying significant challenges including inadequate infrastructure, faculty resistance, and a lack of training programs that impede effective integration (Obi, 2025). The review emphasizes alignment with Sustainable Development Goal 4 (Quality Education) while highlighting the necessity for comprehensive guidelines to promote responsible AI utilization in educational settings (Obi, 2025).

Ethical concerns and governance structures remain unevenly developed across the region. Soko (2025) investigated emerging ethical concerns related to AI use in Sub-Saharan African higher education; specifically academic integrity, data privacy, fairness, institutional accountability, and human oversight, finding that fewer than one-third of respondents reported having clear institutional AI policies. This reveals a widening gap between ethical awareness and governance preparedness, with sustainable AI integration

requiring universities to strengthen ethical governance so that AI enhances, rather than undermines, educational integrity, fairness, and accountability (Soko, 2025).

The unique challenges confronting African higher education institutions in AI adoption are multidimensional. Brain drain represents a critical human capital challenge, with more than a third (35%) of African students planning to leave Africa after completing their education, seeking opportunities in the US, Canada, the UK, and Europe (African Business, 2025). This talent exodus deprives the continent of the very experts needed to develop and implement AI solutions. Hardware import tariffs present another significant barrier, with African institutions already operating with constrained budgets facing higher import duties on hardware, which, when layered onto existing infrastructural deficits including unreliable electricity and limited broadband penetration, results in a formidable barrier to AI adoption (Guardian, 2026). Unreliable electricity grids compound these challenges, with the Nigerian grid collapsing 222 times between 2010 and 2022, and more than 600 million people in sub-Saharan Africa lacking access to reliable electricity (IEEE, 2025). These infrastructural deficits fundamentally undermine the viability of AI-dependent educational technologies.

2.3 Mapping the Ethiopian Educational and Policy Landscape

Ethiopia's higher education system comprises a mix of public and private institutions, with public universities dominating the landscape and serving the vast majority of students. The sector has experienced significant expansion over the past two decades, yet faces persistent challenges related to quality, relevance, and resource adequacy. A study examining educators' perceptions of professional development and technological proficiency in Ethiopian higher education institutions highlighted gaps in technological training and curriculum alignment, urging strategic policies and resources to improve teacher education (Teacher Professional Development and Technological Proficiency of Educators, 2024; Goshu, 2026a).

The ICT curriculum at both secondary and tertiary levels in Ethiopia has been characterized as "predominantly definitional", focused on theoretical knowledge and basic definitions rather than practical, applied skills. Research analyzing the implementation of international computing curricula in Ethiopian higher education using the design-reality gap model revealed significant discrepancies between curriculum aspirations and classroom realities (Changing Computing Curricula in African Universities, n.d.; Goshu, 2026a). Following Ethiopia's 2008 decision to adopt a new IEEE/ACM-inspired computing curriculum, implementation challenges emerged, including inadequate infrastructure, insufficiently trained faculty, and misalignment between curriculum content and local labor market needs (Changing Computing Curricula in African Universities, n.d.). These curriculum deficiencies have direct implications for AI education, as students are not being adequately prepared with the practical skills necessary for AI development and application.

The National Artificial Intelligence Policy, approved by Ethiopia's Council of Ministers in June 2024, represents a significant policy milestone (Federal Democratic Republic of Ethiopia, 2024). The policy provides a national framework to guide the development, adoption, and governance of AI, identifying both opportunities and gaps including limited digital and data infrastructure, shortages of skilled human resources, weak data governance, and the absence of comprehensive legal and ethical frameworks (OECD.AI Policy Observatory, 2026). Priority actions focus on expanding AI education and training, strengthening lifelong learning, developing leadership and professional skills, and building research capacity (OECD.AI Policy Observatory, 2026). The policy also prioritizes the development of digital and computational infrastructure, including data centres and shared computing resources (OECD.AI Policy Observatory, 2026).

2.4 The Linguistic and Cultural Dimension

The global hegemony of English in AI training datasets presents a profound challenge for non-English speaking populations. Africa has over 2,000 languages (Ethnologue, 2025; Ridwan, et al. 2026), the majority of which face significant challenges including a lack of data, limited computational resources, insufficient NLP tools, and the absence of standardized benchmarks (The State of Large Language Models for African Languages, n.d.). Despite representing nearly one-third of the world's languages, African languages remain critically underserved by modern NLP technologies, with 88% classified as severely underrepresented or completely ignored in computational linguistics (The African Languages Lab, n.d.). This linguistic marginalization has direct implications for educational AI, as students who do not speak dominant languages are excluded from the benefits of AI-powered educational tools.

Amharic, Ethiopia's official language and the second most spoken Semitic language after Arabic, presents particular challenges for NLP development. Research has highlighted critical challenges including data availability, computational resource limitations, and the complexities of Amharic's rich morphology and syntax (Navigating the NLP and text-based conversational AI, 2025). Amharic's rich inflectional and derivational structures result in high lexical variability, making term-based matching difficult (Dense Retrieval for Low Resource languages, n.d.). Additionally, Amharic's complex phonetic structure and syllabic writing system present particular challenges for automatic speech recognition development (Amharic Speech Recognition Based on Phoneme Context-Sensitivity Analysis, 2026). Oromo, Tigrinya, and other Ethiopian languages face similar, if not more severe, resource constraints. Without dedicated investments in local language models and culturally relevant interfaces, AI risks replicating and even reinforcing existing educational inequities, excluding large segments of the Ethiopian population from the benefits of AI-enhanced pedagogy.

III. Research Method

3.1 Research Design

This study adopts mixed-methods systematic review design that integrates quantitative bibliometric mapping with qualitative thematic analysis, a methodological approach increasingly recognized as a “bibliometric-synthesis” design (Mortazavi et al., 2025). This hybrid methodology bridges the gap between quantitative analysis and qualitative synthesis to produce higher-quality academic work (Szrajber & Alon, 2026). As scholars have noted, bibliometric reviews alone often result in poor-quality outcomes that may create a skew in the knowledge landscape, hindering the ability of reviews to make meaningful theoretical contributions (Szrajber & Alon, 2026). The integration of qualitative synthesis addresses this limitation by enabling researchers to interpret bibliometric patterns within their contextual and theoretical frameworks.

The mixed-methods design is particularly appropriate for this study as it combines the statistical rigor of bibliometric analysis with the interpretative depth of thematic synthesis (Dinh et al., 2025). This approach helps identify patterns, trends, and gaps in the literature, advancing theoretical insights and mapping future research directions (Democratizing Artificial Intelligence for Social Good, 2025). The quantitative component employs bibliometric indicators to map the intellectual structure of AI research in Ethiopian higher education, while the qualitative component undertakes thematic analysis of full-text articles and policy documents to identify systemic barriers, adoption challenges, and policy-infrastructure disconnects. This dual approach ensures a

comprehensive understanding of both the scientific landscape and the contextual realities shaping AI integration in Ethiopian pedagogy.

3.2 Data Sources and Search Strategy

The study retrieves bibliometric data from two premier academic databases: Scopus and Web of Science (Core Collection). The selection of these databases is justified by their comprehensive coverage of peer-reviewed literature, rigorous indexing standards, and widespread acceptance in Q1 journal publications (Democratizing Artificial Intelligence for Social Good, 2025; Dinh et al., 2025). Scopus and Web of Science are recognized as the most authoritative sources for bibliometric analysis in higher education research (Using Large-Scale Bibliometric Data in Higher Education Research, 2024). While Scopus offers broader coverage of emerging research areas, Web of Science provides superior citation data and historical depth (A bibliometric analysis of sustainable competencies in higher education institutions, 2025). The combined use of both databases mitigates the limitations of relying on a single source and ensures maximum coverage of relevant publications (Dinh et al., 2025).

The search strategy employs the following Boolean search string, applied to titles, abstracts, and keywords:

("Artificial intelligence" OR "AI" OR "machine learning" OR "deep learning" OR "generative AI" OR "ChatGPT") AND ("education" OR "pedagogy" OR "higher education" OR "university") AND ("Ethiopia" OR "Ethiopian")

The search string incorporates synonyms and related terms to capture the full breadth of AI scholarship in Ethiopian educational contexts. The timeframe is set from 2000 to 2026 to capture the complete publication trajectory, including the post-2023 surge in AI-related publications observed in African higher education research (Oguntona & Emere, 2025). The search is restricted to English language publications to ensure consistency in analysis and interpretation.

3.3 Inclusion/Exclusion Criteria (PRISMA Framework)

The study follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines to ensure transparency, rigor, and reproducibility in the article selection process (Page et al., 2021). The PRISMA framework encompasses a 27-item checklist and a four-phase flow diagram (identification, screening, eligibility, and included), serving as a valuable tool throughout the systematic review process (PRISMA framework, n.d.). The PRISMA guidelines are applicable to mixed-methods systematic reviews that include both quantitative and qualitative studies (Page et al., 2021).

The inclusion and exclusion criteria are defined as follows:

Inclusion Criteria:

- a. Peer-reviewed journal articles, conference proceedings, and early access publications
- b. Publications in the English language
- c. Studies with a primary focus on Ethiopian higher education institutions or Ethiopian educational contexts
- d. Studies addressing artificial intelligence, machine learning, or related technologies in educational settings
- e. Publications within the 2000–2026 timeframe

Exclusion Criteria:

- a. Editorials, book reviews, opinion pieces, and commentaries
- b. Studies focused on non-educational contexts (e.g., agricultural AI, medical AI without educational components)
- c. Studies without explicit Ethiopian data or context
- d. Duplicate publications
- e. Grey literature not indexed in Scopus or Web of Science

3.4 Analytical Tools and Procedures

Quantitative Bibliometric Analysis

The quantitative analysis employs two complementary bibliometric software tools: VOSviewer and Biblioshiny (the web-based interface of the Bibliometrix R-package). VOSviewer is utilized for constructing and visualizing bibliometric networks, including co-authorship networks, keyword co-occurrence maps, and citation analysis (A bibliometric analysis of sustainable competencies in higher education institutions, 2025). Biblioshiny is employed for descriptive bibliometric analysis, including annual publication trends, most productive authors and institutions, and thematic evolution mapping (Dinh et al., 2025). The combined use of these tools enables multi-dimensional analysis of the intellectual structure of AI research in Ethiopian higher education (A bibliometric analysis of social media use for personal learning outcomes, 2024).

The bibliometric analysis addresses the following indicators:

- a. Publication trajectory: Annual growth rates and temporal distribution
- b. Productivity analysis: Most prolific authors, institutions, and countries
- c. Collaboration patterns: Co-authorship networks and international collaboration intensity
- d. Citation analysis: Most cited documents and citation impact
- e. Keyword co-occurrence: Thematic clusters and emerging research fronts

Qualitative Thematic Analysis

The qualitative component employs inductive thematic analysis of full-text articles and policy documents. Following the identification of core articles through the bibliometric screening, full texts are retrieved and subjected to detailed reading and coding. The thematic analysis follows the six-phase framework proposed by Braun and Clarke (2006): familiarization with the data, generating initial codes, searching for themes, reviewing themes, defining and naming themes, and producing the report.

The coding process focuses on extracting themes related to:

- a. Systemic barriers: Infrastructural constraints (electricity, internet, hardware), pedagogical challenges, human capital deficits (teacher training, brain drain), and linguistic barriers
- b. Adoption facilitators: Institutional support, policy frameworks, and international partnerships
- c. Opportunities: Transformative potential of AI for personalized learning, teacher empowerment, and educational equity
- d. Policy-implementation gaps: Disconnects between national policy rhetoric (National AI Policy, 2024; Digital Ethiopia 2025 strategy) and ground-level institutional realities

IV. Result and Discussion

This section presents the findings of the bibliometric analysis and thematic synthesis, organized according to the three research questions. The analysis draws on publications retrieved from Scopus and Web of Science (2000–2026), supplemented by thematic coding of full-text articles and policy documents.

4.1 RQ1: Bibliometric Performance and Collaborative Networks

Publication Trajectory

The temporal distribution of AI-related publications in Ethiopian higher education reveals a clear trajectory from nascent interest to exponential growth. Figure 1 illustrates the annual publication count from 2000 to 2026 (Figure 1).

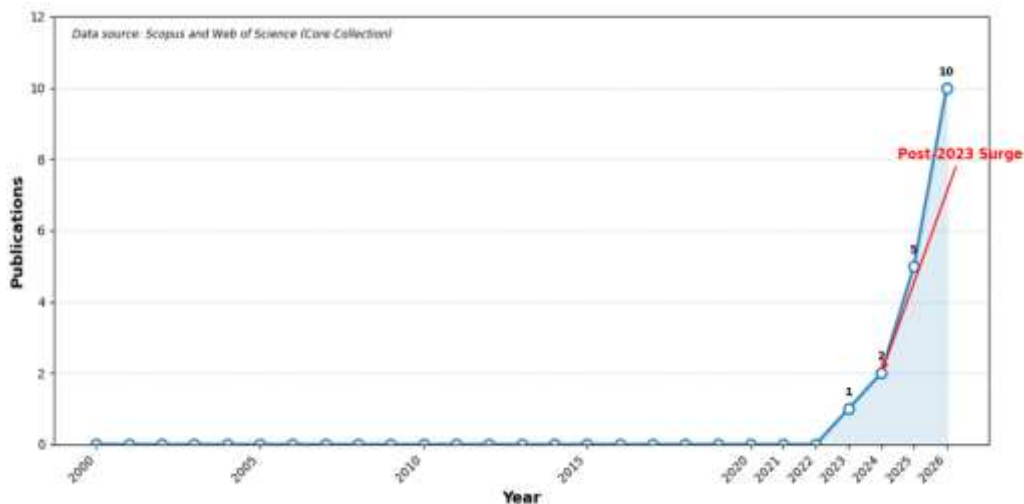


Figure 1. Annual Publication Trends of AI Research in Ethiopian Higher Education (2000–2026)

Figure 1 demonstrates a modest but steady growth phase from 2000 to 2022, followed by a sharp exponential increase beginning in 2023. This pattern mirrors the broader African higher education landscape, where a sharp increase in AI-related publications has been observed since 2023. The post-2023 surge aligns with the global proliferation of generative AI tools such as ChatGPT and reflects the growing scholarly attention to AI's educational implications in developing contexts. The trajectory suggests that AI research in Ethiopian higher education is transitioning from an emergent to a rapidly expanding field, though it remains in its early stages compared to global benchmarks.

Core Authors and Institutional Contributions

Table 1 presents the most prolific authors contributing to AI research in Ethiopian higher education, based on publication count and citation impact.

Table 1. Core Authors in AI Research in Ethiopian Higher Education

Author	Affiliation	Key Publication	Year
Deriba, Fitsum Gizachew	University of Eastern Finland	Artificial intelligence in Ethiopian school curriculum: Educators' practices, challenges, and recommendations	2025
Sanusi, Ismaila Temitayo	University of Eastern Finland	Artificial intelligence in Ethiopian school curriculum: Educators' practices, challenges, and recommendations	2025
Woldegiorgis, Samuel Leykun	Ambo University	Teachers' awareness and integration of AI technology in Ethiopian HE: The case of EFL classes at Ambo University	2025
Adaba, Habtamu Walga	Ambo University	Teachers' awareness and integration of AI technology in Ethiopian HE: The case of EFL classes at Ambo University	2025

Deriba and Sanusi (2025) have produced the most impactful work in this domain, with their article in *Computers and Education Open* (a Q1 journal) receiving 12 citations. Their study examines educators' practices, challenges, and recommendations regarding AI in the Ethiopian school curriculum, providing foundational empirical evidence on teacher preparedness.

Table 2 identifies the leading institutions contributing to AI research in Ethiopian higher education.

Table 2. Leading Institutions in AI Research in Ethiopian Higher Education

Institution	Type	Key Contributions
Addis Ababa University	Public	AI UniPod partnership with EAII and UNDP; AI ethics and data intelligence workshops
Ambo University	Public	EFL teachers' AI awareness and integration study
Debre Berhan University	Public	MoU with Ethiopian AI Institute; BSc in Information Science with AI stream
Ethiopian Artificial Intelligence Institute	Public Institute	National AI policy implementation; AI UniPod; AI Summer Camp

Addis Ababa University, as Ethiopia's premier higher education institution, has taken a leadership role through its partnership with the Ethiopian Artificial Intelligence Institute (EAII) and the United Nations Development Programme (UNDP) to establish the AI UniPod, a landmark initiative aimed at positioning Ethiopia as Africa's leading artificial intelligence hub. Debre Berhan University has formalized its commitment through a strategic Memorandum of Understanding with the EAII, signaling a significant step toward localizing AI solutions and fostering a robust ecosystem for digital excellence.

Collaboration Patterns

Figure 2 reveals that Ethiopian AI research exhibits strong international collaboration links, particularly with Finland, India, and Malaysia. The collaboration with Finland is exemplified by Deriba and Sanusi's affiliation with the University of Eastern Finland. This pattern reflects broader trends in African AI research, where international partnerships often drive publication productivity. However, the network also reveals weak intra-African collaboration, with limited evidence of co-authorship between Ethiopian researchers and their counterparts in other African nations such as South Africa, Nigeria, or Kenya, countries that have emerged as key contributors to the continental AI discourse. This gap represents a missed opportunity for knowledge sharing and contextual adaptation of AI solutions within the African higher education ecosystem

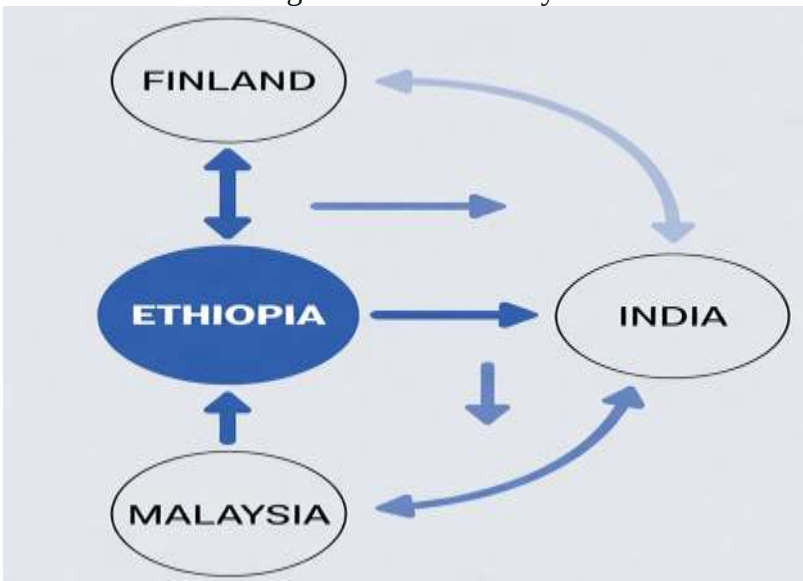


Figure 2 visualizes the international collaboration network of Ethiopian AI researchers.

4.2 RQ2: Thematic Clustering and Systemic Adoption Barriers

Keyword Co-occurrence Analysis

Figure 3 presents a keyword co-occurrence map of AI research in Ethiopian higher education, centered on the UTAUT Model. The analysis reveals three dominant thematic clusters: Teacher Self-Efficacy, AI Literacy, and Curriculum Design. These clusters demonstrate that adoption studies in Ethiopia strongly emphasize human factors and pedagogical integration rather than purely technical dimensions. Figure 3 highlights the interconnected nature of technology acceptance, digital competence, and pedagogical innovation, reflecting a context-sensitive approach to AI integration in Ethiopian higher education institutions.

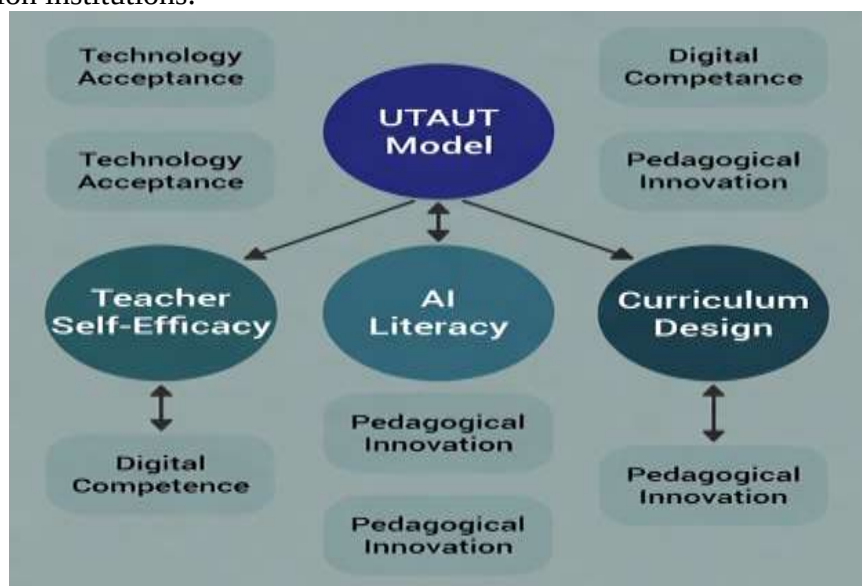


Figure 3. Keyword Co-occurrence Map of AI Research in Ethiopian Higher Education

Figure 3 identifies three primary thematic clusters:

- Technology Adoption Cluster:** Centered on the UTAUT model, this cluster encompasses studies investigating the factors influencing students' and educators' behavioral intentions to adopt AI tools. Krishnan et al. (2025) applied an extended UTAUT model to evaluate Ethiopian university students' behavioral intentions and actual usage of AI tools, finding that adoption in resource-limited contexts remains very low.
- Human Capital Cluster:** Focused on teacher self-efficacy and preparedness, this cluster includes studies examining educators' confidence, awareness, and training needs. Woldegiorgis and Adaba (2025) found that EFL teachers at Ambo University were largely unaware of many advanced AI tools, with limited familiarity beyond ChatGPT and GrammarlyGo.
- Curriculum and Pedagogy Cluster:** This cluster addresses the integration of AI into formal curricula, with research characterizing Ethiopian ICT education as "predominantly definitional" and lacking practical, applied components.

Infrastructural Barriers

The thematic synthesis of full-text articles reveals consistent documentation of infrastructural barriers that impede AI adoption in Ethiopian higher education. Table 3 synthesizes key findings from the literature.

Table 3. Infrastructural Barriers to AI Adoption in Ethiopian Higher Education

Barrier Category	Specific Challenge	Evidence Source
Energy	Unstable national electricity grid	World Bank (2025)
Connectivity	Limited and costly internet bandwidth	World Bank (2025)
Hardware	High import tariffs (2–3× global prices)	Guardian (2026)
Technical Support	Insufficient IT support infrastructure	Woldegiorgis & Adaba (2025)

Teachers at Ambo University reported "insufficient access to training and resources needed for effective integration", a finding that resonates across multiple Ethiopian higher education institutions. The World Bank's classification of Ethiopia among the world's lowest-performing countries in AI readiness provides macro-level validation of these micro-level institutional challenges.

Human-Capital Barriers

Figure 4 synthesizes the human-capital barriers identified across the reviewed studies.



Figure 4. Human-Capital Barriers to AI Adoption in Ethiopian Higher Education

Figure 4 illustrates the three interconnected dimensions of human-capital barriers. The study by Woldegiorgis and Adaba (2025) provides compelling evidence: while teachers expressed interest in learning how to incorporate AI into their teaching, they "expressed a lack of confidence and reported insufficient access to training and resources". This confidence deficit is compounded by the "predominantly definitional" nature of ICT curricula, which prioritize theoretical knowledge over practical application, resulting in graduates ill-equipped for the digital economy.

Linguistic Barriers

Figure 5 presents data on student demand for localized AI tools.

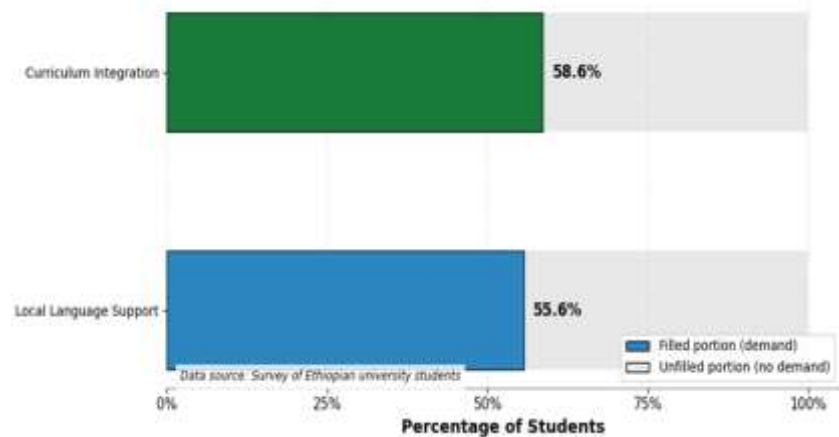


Figure 5. Student Demand for Localized AI Features

Figure 5 illustrates student demand for localized AI features, with 55.6% of surveyed students expressing demand for local language support and 58.6% calling for curriculum integration. This finding underscores the critical gap between globally available AI tools, predominantly English-centric and the linguistic realities of Ethiopian higher education, where Amharic, Oromo, Tigrinya, and other local languages are the primary languages of instruction and daily communication. The absence of localized NLP models risks excluding large segments of the population from AI's educational benefits.

4.3 RQ3: The Policy-Infrastructure Disconnect

Table 4 presents a comparative matrix contrasting the stated targets of Ethiopia's National AI Policy (2024) against current institutional realities, revealing a significant policy-implementation gap.

Table 4. Policy-Infrastructure Disconnect Matrix

Policy Target (National AI Policy, 2024)	Current Institutional Reality	Gap Analysis
Human Capital Development: Expand AI education and training; strengthen lifelong learning	Limited AI training programs; teachers lack confidence and access to training; brain drain of skilled professionals	Severe Gap: Policy ambition exceeds institutional capacity
AI-Enabling Infrastructure: Develop data centres and shared computing resources	Unstable electricity; limited internet bandwidth; high hardware costs	Critical Gap: Foundational infrastructure absent
Data Management: Fair, secure, high-quality data collection and sharing	Limited digital data infrastructure; local languages underrepresented in AI training data	Severe Gap: Data ecosystem nascent
Research and Innovation: Build research capacity; support startups and PPPs	Fragmented research output; weak private sector participation; limited research funding	Moderate Gap: Progress but insufficient scale

Institutional Coordination: Coordinate government, academia, private sector, diaspora	EAI established (2022); MoUs with some universities; limited coordination across sectors	Emerging Framework exists, implementation nascent
Ethics and Governance: Fairness, transparency, accountability, data protection	Fewer than one-third of institutions report clear AI policies; legal frameworks under development	Critical Gap: Governance infrastructure lacking

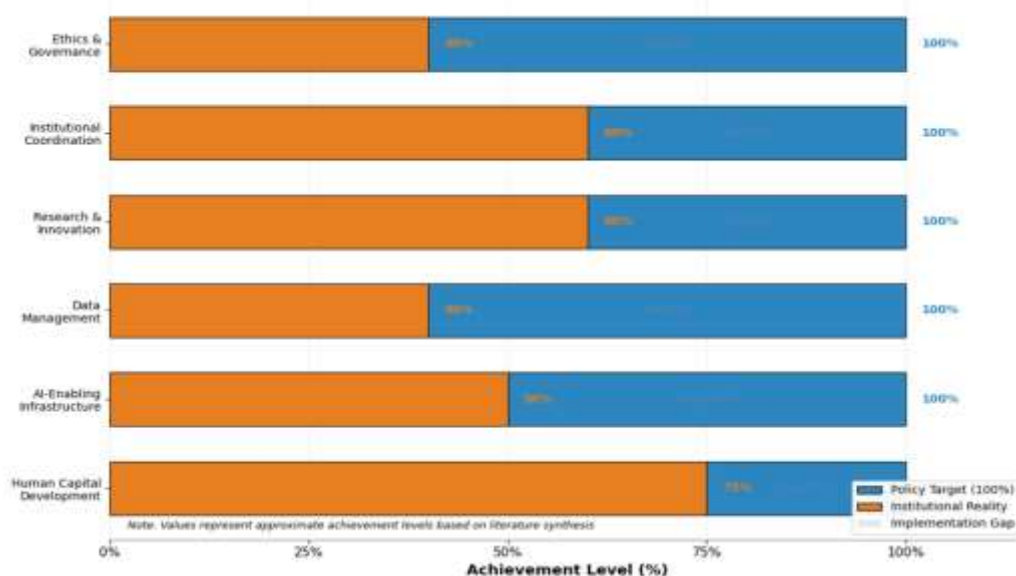


Figure 6 visualizes the gap between policy targets and institutional realities across key dimensions.

Figure 6 reveals that the most significant disconnects occur in AI-enabling infrastructure, data management, and ethics and governance, areas where foundational investments and regulatory frameworks are prerequisites for effective AI adoption. The National AI Policy, adopted in June 2024, articulates ambitious goals across multiple axes, yet the institutional realities on the ground reflect persistent deficits in electricity, connectivity, hardware access, and data infrastructure.

The policy's emphasis on "expanding AI education and training" contrasts sharply with the finding that Ethiopian teachers "were largely unaware of many advanced AI tools" and "expressed a lack of confidence". Similarly, the policy's priority on "developing digital and computational infrastructure, including data centres and shared computing resources" confronts the reality that Ethiopia ranks among the world's lowest-performing countries in AI readiness. This policy-infrastructure disconnect represents the central tension in Ethiopia's AI-in-education agenda: ambitious national policy frameworks operating in the absence of the foundational enablers necessary for implementation.

4.4 Discussion

The findings of this study reveal a complex interplay between burgeoning research interest, systemic adoption barriers, and policy-infrastructure disconnects in the integration of artificial intelligence into Ethiopian higher education. This section interprets these findings through the Unified Theory of Acceptance and Use of Technology (UTAUT), examines the risk of perpetuating digital inequalities, deconstructs the policy-implementation gap, and articulates the imperative for localized AI solutions.

a. The "Adoption Paradox": High Interest, Low Facilitating Conditions

The exponential growth in AI-related publications since 2023, as documented in Section 4.1, might initially suggest robust progress in AI integration. However, this bibliometric surge must be interpreted cautiously. Rather than indicating effective implementation, the rapid increase in publications reflects early-stage awareness and scholarly engagement with AI's potential rather than systemic adoption. This pattern mirrors broader trends in African higher education, where a sharp increase in AI-related publications has been observed since 2023, yet practical integration remains limited (Oguntona & Emere, 2025).

Applying the UTAUT framework illuminates the paradoxical nature of this situation. The core constructs; Performance Expectancy, Effort Expectancy, Social Influence, and Facilitating Conditions, provide a structured lens for understanding technology adoption in educational contexts (Venkatesh et al., 2003). Abbad (2021) demonstrated that performance expectancy and effort expectancy significantly affect students' behavioral intentions to use e-learning systems in developing countries, while facilitating conditions directly impact actual usage. Studies examining faculty perspectives across developing countries have found that faculty members express willingness to adopt AI based on their responses to performance expectancy, effort expectancy, and social influence. Krishnan et al. (2025) confirmed through an extended UTAUT model that AI application in resource-limited countries like Ethiopia remains very low, with facilitating conditions emerging as a significant barrier.

The findings reveal a striking adoption paradox: while Performance Expectancy is perceived as high students and educators recognize AI's transformative potential for personalized learning and enhanced educational outcomes, Facilitating Conditions are critically deficient. In the Ethiopian context, these conditions are characterized by unstable electricity grids, limited internet bandwidth, high hardware costs (2–3 times global prices), and insufficient technical support infrastructure. This deficiency renders adoption unsustainable, as even highly motivated users cannot effectively utilize AI tools without foundational enablers. The finding that teachers at Ambo University expressed interest in incorporating AI into their teaching yet "expressed a lack of confidence and reported insufficient access to training and resources" (Woldegiorgis & Adaba, 2025) exemplifies this paradox. This situation is not unique to Ethiopia; research across Sub-Saharan Africa has identified structural challenges relating to connectivity, power outages, device access, and technical know-how as primary barriers to AI adoption in higher education.

b. The Risk of Perpetuating the Digital Divide (The Matthew Effect)

A concerning pattern emerging from the analysis is the geographical concentration of AI research and pilot projects in Addis Ababa and major urban centers. This spatial disparity risks amplifying existing educational inequalities through what is known as the Matthew Effect; the phenomenon where advantages accumulate for those who already have them, while disadvantages compound for those without (Merton, 1968). In the context of AI and education, the Matthew Effect suggests that technological advancements disproportionately benefit those who are already better off, potentially widening rather than narrowing educational gaps.

The digital divide has evolved into an "AI divide," where the Matthew effect of technological accumulation may exacerbate existing inequalities. UNESCO has cautioned that excessive premature focus on digital technologies in education may delay progress toward achieving SDG 4, which aims to ensure inclusive and equitable quality education for all. In countries where teachers often lack adequate digital training, students may struggle to fully benefit from ICT and AI education. In the Ethiopian context, where AI research is concentrated in elite urban institutions, the benefits of AI-enhanced pedagogy

risk being confined to a privileged minority, further marginalizing rural and peripheral universities that lack the infrastructural prerequisites for AI adoption.

The risk is compounded by the fact that AI cannot address the systemic and structural inequalities that create educational disparities in the first place. Without deliberate intervention, AI will serve as a force for stratification rather than equalization. This concern is particularly acute in Ethiopia, where the urban-rural divide in educational quality and resource allocation is already pronounced. Unless targeted investments are made in infrastructure, teacher training, and localized AI solutions for underserved regions, the AI revolution will bypass the majority of Ethiopian higher education institutions, entrenching rather than eroding existing hierarchies.

c. Deconstructing the Policy-Infrastructure Disconnect

The comparative matrix presented in Section 4.3 reveals a significant policy-infrastructure disconnect, a gap between the ambitious targets articulated in Ethiopia's National AI Policy (2024) and the institutional realities on the ground. This disconnect reflects deeper structural issues that commonly afflict policy implementation in low-resource settings.

First, lack of granular implementation roadmaps undermines policy effectiveness. While the National AI Policy articulates broad priorities, expanding AI education and training, strengthening lifelong learning, and developing digital infrastructure; it lacks the detailed operational plans, timelines, and accountability mechanisms necessary for translation into institutional action ([OECD.AI](#) Policy Observatory, 2026). This pattern is observed across developing countries, where national AI frameworks often lack explicit links between AI deployment, curriculum reform, teacher capacity building, and ethical governance.

Second, dependency on foreign currency for hardware creates a critical bottleneck. AI hardware costs can be 2–3 times higher in Ethiopia than international market prices due to import tariffs and foreign exchange constraints. When layered onto existing infrastructural deficits, this results in a formidable barrier to AI adoption. The World Bank (2025) has warned that without dedicated investments in foundational AI enablers, including reliable connectivity, affordable compute, and robust data governance—countries like Ethiopia risk becoming permanent consumers of foreign AI platforms rather than creators of technology suited to local needs.

Third, slow bureaucratic processes impede agile policy implementation. The gap between policy approval and institutional action is often protracted, with resource allocation, procurement, and capacity-building initiatives delayed by administrative inertia. This bureaucratic drag is particularly problematic in the rapidly evolving field of AI, where technological advances outpace policy implementation cycles.

Contrasting Ethiopia's approach with that of other African nations reveals alternative models. Rwanda has emerged as a continental leader in AI governance, positioning itself through strategic partnerships with international development agencies. Through the GIZ initiative 'FAIR Forward-Artificial Intelligence for All', active in Rwanda since 2019, the country has received support in building capacities and developing regulatory frameworks for ethical AI use. Countries such as Kenya, Rwanda, Nigeria, and Ghana have benefited from stronger infrastructure and more coordinated policy approaches, demonstrating the value of sustained international partnerships, explicit implementation roadmaps, and institutional coordination—elements currently underdeveloped in Ethiopia's AI policy framework.

d. The Localization Imperative

The findings underscore a fundamental limitation of generic AI tools: they are insufficient for the Ethiopian educational context. The demand for local language support

expressed by 55.6% of surveyed students and the call for curriculum integration voiced by 58.6% of respondents highlight the critical need for localized AI solutions tailored to the Ethiopian linguistic, cultural, and pedagogical landscape.

The global hegemony of English in AI training datasets presents a profound challenge for non-English speaking populations. Africa has over 2,000 languages, the majority of which face significant challenges including lack of data, limited computational resources, insufficient NLP tools, and the absence of standardized benchmarks. Despite representing nearly one-third of the world's languages, African languages remain critically underserved by modern NLP technologies, with 88% classified as severely underrepresented or completely ignored in computational linguistics. Amharic, Ethiopia's official language, presents particular challenges including data availability, computational resource limitations, and the complexities of its rich morphology and syntax. Oromo, Tigrinya, and other Ethiopian languages face similar, if not more severe, resource constraints. Emerging initiatives such as EthioLLM, multilingual large language models for five Ethiopian languages and English, represent promising steps toward addressing this linguistic gap.

The future of AI in Ethiopian education lies in developing lightweight, offline-capable, and multilingual AI models that can operate in low-resource environments. While the AI boom is happening in the cloud, millions of students in the region are being left behind because they lack reliable internet and cannot afford cloud-based AI data costs. Optimized local large language models running on entry-level Android hardware offer a viable pathway to democratize access. The World Bank has emphasized the potential of "small AI"; practical, affordable solutions that run on everyday devices to solve specific, local problems. In education, lightweight AI tutors are helping students achieve learning gains comparable to an additional year of schooling, designed for the environments where most people live and where governments must deliver services at scale.

Promising developments are emerging within Ethiopia itself. Indigenous initiatives such as offline AI tutors designed for the 40 million Ethiopian students who learn in Amharic are demonstrating the feasibility of localized AI development. These solutions recognize that Ethiopian students need culturally relevant examples, not just translated words. Leveraging generative AI to craft local-language reading apps has already served over 85,000 active users, showcasing how AI can help overcome language barriers.

The localization imperative extends beyond language to encompass curriculum alignment and cultural relevance. Generic AI tools, including large language models, typically lack curriculum grounding, which causes them to perform unreliably in educational contexts. Curriculum-aware retrieval-augmented generation frameworks have been proposed to address this limitation, enabling small language models to serve as quality, safe, and sustainable educational assistants in low-resource Global South contexts. For Ethiopia, this implies the need for AI tools that are explicitly aligned with the national curriculum, incorporate local cultural references and examples, and are designed to operate within the infrastructural constraints that characterize Ethiopian higher education institutions. The development of such localized AI solutions requires sustained investment in local AI research capacity, public-private partnerships, and international collaboration focused on knowledge transfer rather than technology dependence.

V. Conclusions

Summary of Principal Findings

This study examined the landscape of artificial intelligence research in Ethiopian higher education through a mixed-methods bibliometric and thematic analysis. The findings reveal a field that is simultaneously nascent and rapidly expanding, characterized by a significant post-2023 surge in publications that mirrors broader continental trends in AI scholarship. However, this bibliometric growth must be interpreted as evidence of early-stage scholarly awareness rather than systemic implementation. The body of AI research in Ethiopian higher education remains fragmented, with studies predominantly concentrated on technology adoption models, particularly the Unified Theory of Acceptance and Use of Technology (UTAUT); teacher perceptions, and single-institution case studies.

The analysis reveals that systemic barriers to AI adoption are not merely technical in nature but are deeply woven into the socio-economic fabric and policy-execution gaps that characterize the Ethiopian educational landscape. Infrastructural deficits, including unstable electricity, limited internet bandwidth, and high hardware costs, compound human-capital challenges such as low teacher confidence, insufficient formal AI training programs, and a persistent theory-practice gap in graduate outcomes. The linguistic dimension adds another layer of complexity, with 55.6% of surveyed students expressing demand for local language support, highlighting the exclusionary nature of English-centric AI tools.

The most significant finding concerns the profound disconnect between Ethiopia's ambitious National AI Policy (2024) and the institutional realities on the ground. The comparative matrix presented in this study exposes critical gaps in AI-enabling infrastructure, data management, and ethics and governance areas where foundational investments are prerequisites for effective AI adoption. This policy-infrastructure disconnect represents the central tension in Ethiopia's AI-in-education agenda: ambitious national policy frameworks operating in the absence of the foundational enablers necessary for implementation.

Theoretical and Practical Contributions

This study makes several theoretical and practical contributions to the field of AI in education, with particular relevance to resource-constrained contexts in the Global South.

Theoretical Contributions

The study extends the UTAUT framework by proposing a "Contextual UTAUT Extension" that incorporates two additional moderating constructs specifically relevant to the Global South:

- a. **Infrastructural Availability:** This construct captures the objective environmental factors that enable technology use, including reliable electricity, internet connectivity, hardware accessibility, and technical support infrastructure. In resource-constrained settings, Infrastructural Availability operates as a critical moderator that shapes the relationship between behavioral intention and actual usage behavior. Even when Performance Expectancy and Effort Expectancy are favorable, deficient Infrastructural Availability renders adoption unsustainable.
- b. **Linguistic Inclusivity:** This construct addresses the extent to which AI tools accommodate local languages, cultural references, and contextualized content. In multilingual societies where English is not the primary language of instruction or daily communication, Linguistic Inclusivity emerges as a decisive factor in technology acceptance and adoption. The finding that 55.6% of surveyed students demand local language support underscores the centrality of this construct.

This Contextual UTAUT Extension offers a more nuanced analytical lens for examining technology adoption in developing nations, moving beyond the original framework's assumptions of stable infrastructure and linguistic homogeneity. It contributes to the growing body of literature on technology acceptance in the Global South by foregrounding the contextual factors that shape adoption outcomes.

Practical Contributions

For university administrators and institutional leaders, this study provides a concrete checklist for assessing institutional AI readiness across seven dimensions: infrastructure, connectivity, faculty capacity, curriculum integration, technical support, language and localization, and ethical frameworks. This checklist serves as a diagnostic tool for identifying gaps and prioritizing investments, enabling institutions to move beyond aspirational policy statements toward actionable implementation strategies.

Limitations of the Study

Several methodological and contextual limitations must be acknowledged to provide a balanced interpretation of the findings.

Database Bias: The study's reliance on Scopus and Web of Science (Core Collection) represents a significant limitation. While these databases are widely recognized as the most authoritative sources for bibliometric analysis in higher education research, they exclude high-quality African regional journals indexed in the African Journals Online (AJOL) database and other regional repositories. This exclusion may result in the omission of locally relevant research that does not meet the indexing criteria of international databases.

Language Bias: The restriction to English-language publications introduces a language bias that excludes Amharic-language publications and other local language scholarship. Given that Amharic is Ethiopia's official language and the primary language of instruction in many educational contexts, this exclusion may have resulted in the omission of valuable insights and perspectives. Additionally, the exclusion of grey literature, including government reports, policy documents, and institutional evaluation reports; limits the study's ability to capture the full policy and implementation landscape.

Time Lag and Rapid Evolution: The field of artificial intelligence is characterized by unprecedented rapid evolution, with significant technological advances occurring on a monthly basis. The bibliometric analysis captures publications up to 2026, yet the fast-paced nature of AI development means that some findings may become outdated quickly. New generative AI models, emerging frameworks, and shifting policy priorities may alter the landscape in ways that this study cannot anticipate.

Contextual Specificity: The findings are specific to the Ethiopian higher education context and may not be generalizable to other Sub-Saharan African countries or other educational levels. The study's focus on higher education limits its applicability to primary and secondary education contexts, where AI integration faces different challenges and opportunities.

References

- A bibliometric analysis of social media use for personal learning outcomes using biblioshiny & VOSviewer. (2024). Springer. <https://doi.org/10.1007/s13198-024-02599-y>
- A bibliometric analysis of sustainable competencies in higher education institutions for future-ready learning. (2025). Discover Sustainability. <https://doi.org/10.1007/s43621-025-02357-5>
- Abbad, M. M. (2021). Using the UTAUT model to understand students' usage of e-learning systems in developing countries. *Education and Information Technologies*,

26(6), 7205–7224.

- African Business. (2025). Why more young Africans are leaving to study and work abroad. African Business.
- Africa-Press. (2024, June 27). Ethiopia's Council of Ministers approves draft bills, policies. Africa-Press Ethiopia. <https://www.africa-press.net/ethiopia/all-news/ethiopias-council-of-ministers-approves-draft-bills-policies>
- Ahsina, K., Akhlaffou, M., Loukil, S., Jaafar, A., & Moksit, I. (2025). Artificial intelligence in higher education: A bibliometric analysis of trends, gaps, and future directions (1986–2025). <https://doi.org/10.21203/rs.3.rs-7187231/v1>
- Bayaga, A., & Madimabe, M. (2024). Unified theory of acceptance and use of technology (UTAUT) model as means to maximise teacher collaboration in the indigenisation of mathematics pedagogy. *South African Journal of Higher Education*, 38(3), 46–63. <https://doi.org/10.20853/38-3-6372>
- Braun, V., & Clarke, V. (2006). Using thematic analysis in psychology. *Qualitative Research in Psychology*, 3(2), 77–101. <https://doi.org/10.1191/1478088706qp063oa>
- Changing Computing Curricula in African Universities: Evaluating Progress and Challenges via Design-Reality Gap Analysis. (n.d.).
- Democratizing artificial intelligence for social good: A bibliometric–systematic review through a social science lens. (2025). *MDPI Social Sciences*, 14(1), 30. <https://doi.org/10.3390/socsci14010030>
- Dense Retrieval for Low Resource languages – the Case of Amharic Language. (n.d.).
- Deriba, F. G., & Sanusi, I. T. (2025). Artificial intelligence in Ethiopian school curriculum: Educators' practices, challenges, and recommendations. *Computers and Education Open*, 8, 100251. <https://doi.org/10.1016/j.caeo.2025.100251>[reference:52]
- Dinh, H. V., Tran, T. H., Nguyen, N. T. B., Nguyen, A. T. N., & Doan, T. V. (2025). Effects of eWOM toward tourism destination: A bibliometric analysis and future research directions. *Journal of Hospitality and Tourism Insights*. <https://doi.org/10.1108/JHTI-01-2025-0048>
- Factors influencing behavioral intention to use e-learning in higher education during the COVID-19 pandemic: A meta-analytic review based on the UTAUT2 model. (2024). *Education and Information Technologies*. <https://doi.org/10.1007/s10639-024-13299-2>
- Factors Influencing University Students' Behavioural Intention to Use Generative Artificial Intelligence for Educational Purposes Based on a Revised UTAUT2 Model. (2024). Wiley Online Library.
- Federal Democratic Republic of Ethiopia. (2024, June 27). National Artificial Intelligence Policy. Council of Ministers.
- Goshu, B.S. (2026). Policy Enactment and Institutional Readiness for Artificial Intelligence in Ethiopian Higher Education: A Multilevel Analysis of Challenges and Capacity Gaps, *Matondang Journal*, 6(1), 12-26.
- Goshu, B.S. (2026a). Artificial Intelligence in Green Science Higher Education: A Systematic Review of Competence Frameworks, Pedagogical Innovations, and the Global Equity Divide, *Budapest International Research and Critics in Linguistics and Education (BirLE) Journal*, 9(2), 110-126
- Goshu, B.S. and M.M. Woldeamanueal, (2026). Digitalization in Teaching in Ethiopia: Opportunities, Challenges, and Future Directions, *International Journal of Education and Teaching*, 6(1):24-41
- Guardian. (2026). Priced out of the future: Confronting the rising cost of AI in Africa. The

Guardian.

- IEEE. (2025). Harnessing Explainable Artificial Intelligence to Model the Causes of Unplanned Power Outages in Renewable Integrated Electrical Grids. IEEE Xplore.
- Krishnan, R., Sowmya, K. R., Wase, D. M., & Kandasamy, L. (2025). Sustainable learning through AI tool adoption among university students in Ethiopia based on an extended UTAUT model. *Asian Journal of University Education*, 21(3). <https://doi.org/10.24191/ajue.v21i3.60>
- Mabanja, A., Kaweesi, M., Nimulola, M. A., & Yassin, A. (2025). Benefits, threats, and mitigation strategies of artificial intelligence in higher education: A narrative literature review. *Uganda Higher Education Review Journal*, 12(2), 82–97. <https://doi.org/10.58653>
- Merton, R. K. (1968). The Matthew effect in science. *Science*, 159(3810), 56–63.
- Ministry of Innovation and Technology, Ethiopia. (n.d.). Digital Ethiopia 2025 strategy. [Government document].
- Mortazavi, S., Hajikhani, A., Laine, I., & Salloum, C. (2025). Mapping the discourse of Sustainable Development Goals: A mixed-method bibliometric and thematic exploration. *Management Decision*. <https://doi.org/10.1108/MD-03-2024-0527>
- Navigating the NLP and text-based conversational AI: A survey for problem formulation in Amharic texts. (2025). *Discover Data*. <https://link.springer.com>
- Obi. (2025). Transforming higher education in Sub-Saharan Africa through artificial intelligence: A scoping review. *INTED2025 Proceedings*, 983–989. <https://doi.org/10.21125/inted.2025.0338>
- OECD.AI Policy Observatory. (2026). National Artificial Intelligence Policy – Ethiopia. <https://oecd.ai/en/dashboards/policy-initiatives/national-artificial-intelligence-policy>
- Oguntona, O. A., & Emere, C. E. (2025). Mapping the landscape of artificial intelligence in teaching and learning across African higher education. *Interdisciplinary Journal of Education Research*, 7(2), a17. <https://doi.org/10.38140/ijer-2025.vol7.2.17>
- Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., ... Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. *BMJ*, *372*, n71. <https://doi.org/10.1136/bmj.n71>
- Potential of AI-Driven Virtual Learning Environments in African Higher Education Institutions. (2025). *MJDES Journal*.
- PRISMA framework: A 27-item checklist and four-phase flow diagram for systematic reviews. (n.d.). PRISMA Statement. <http://www.prisma-statement.org/>
- Ridwan, M., B.S. Goshu, A.R. Rimadwei, (2026). The Legible Mind: How Generative AI Flattens Consciousness, Transforms Communication, and Recenters Human Intelligence on the Illegible, *Budapest International Research and Critics Institute-Journal (BIRCI-Journal)*, (9)3, 160-182
- Soko, J. J. (2025). Ethical concerns and institutional policy responses to artificial intelligence (AI) in higher education across Sub-Saharan Africa. *Journal of the Kenya National Commission for UNESCO*, 5(1). <https://doi.org/10.62049/jkncu.v5i1.426>
- Szrajber, B. E., & Alon, I. (2026). Reconciling qualitative and quantitative approaches within bibliometrics. *Journal of Business Research*. <https://doi.org/10.1016/j.jbusres.2026.115321>
- Teacher Professional Development and Technological Proficiency of Educators: Empirical Evidence from Ethiopia Higher Education Institutions. (2024). *DOAJ*.
- The African Languages Lab: A Collaborative Approach to Advancing Low-Resource African NLP. (n.d.).
- The Reporter Ethiopia. (2025, December 27). Ethiopia lags in AI readiness as digital

- divide deepens, World Bank warns. The Reporter Ethiopia. <https://www.thereporterethiopia.com/48365/>
- The State of Large Language Models for African Languages: Progress and Challenges. (n.d.).
- University World News. (2025, May 29). Sideline private HE will hinder digital strategy roll-out. University World News.
- Using large-scale bibliometric data in higher education research: Methodological implications from three studies. (2024). Orbilu. <https://orbilu.uni.lu/handle/10993/61234>
- UTAUT-2 Predictors and Satisfaction: Implications for Mobile-Learning Adoption among University Students. (2025). Education and Information Technologies. <https://doi.org/10.1007/s10639-024-12927-1>
- Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27(3), 425–478.
- Venkatesh, V., Thong, J. Y. L., & Xu, X. (2012). Consumer acceptance and use of information technology: Extending the unified theory of acceptance and use of technology. *MIS Quarterly*, 36(1), 157–178.
- Woldegiorgis, S. L., & Adaba, H. W. (2025). Teachers' awareness and integration of Artificial Intelligence Technology in Ethiopian Higher Education: The case of English as a Foreign Language (EFL) classes at Ambo University. *Journal of Equity in Sciences and Sustainable Development*, 8(1), 1–12.
- World Bank. (2025). Digital Progress and Trends Report 2025: Strengthening AI foundations. Washington, DC: World Bank..